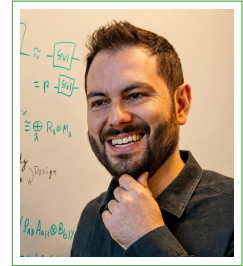


Marco Túlio Quintino

Curriculum Vitae

Associate Professor
(Maître de Conférences, HDR)
Sorbonne University, LIP6
Paris, France

✉ Marco.Quintino@lip6.fr
🌐 mtcq.github.io
Google Scholar / arXiv



Last updated: 23 August 2026

Research interests

Quantum information and computation, quantum correlations (entanglement, Bell nonlocality, EPR steering, and measurement incompatibility), quantum state and channel discrimination, higher-order quantum operations, causality in quantum theory, and semidefinite optimisation.

Current employment

Sep 2022 – **Associate Professor (Maître de Conférences, HDR)**, Sorbonne University, Paris, France
Present
Member of the [Quantum Information](#) team at [LIP6](#)

Employment history

Mar 2021 – **Postdoctoral researcher**, IQOQI Vienna, Vienna, Austria
Aug 2022 Postdoctoral researcher in the group of [Prof. Časlav Brukner](#)
Funding: [ESQ postdoc fellowship](#)
Jun 2020 – **Postdoctoral researcher**, University of Vienna, Vienna, Austria
Mar 2021 Postdoctoral researcher in the group of [Prof. Časlav Brukner](#)
Funding: [Austrian Science Fund \(FWF\)](#) through the [SFB project BeyondC](#)
Mar 2019 – **Postdoctoral researcher**, The University of Tokyo, Tokyo, Japan
May 2020 Postdoctoral researcher in the group of [Prof. Mio Murao](#)
Funding: [Q-LEAP project of the MEXT Japan](#)
Nov 2016 – **Postdoctoral researcher**, The University of Tokyo, Tokyo, Japan
Nov 2018 Postdoctoral researcher in the group of [Prof. Mio Murao](#)
Funding: [JSPS postdoc fellowship](#)

Education

Jan 2025 **Habilitation in Computer Science [HDR]**, Sorbonne University, Paris, France
Thesis: [Quantum information processing via higher-order operations](#)
Defence date: 17 Jan 2025
Jury: [Pablo Arrighi](#), [Anne Broadbent](#), [Giulio Chiribella](#), [Elham Kashefi](#), [Mário Ziman](#)
Oct 2012 – **PhD in Physics**, Université de Genève, Geneva, Switzerland
Sep 2016 Thesis: [Quantum entanglement and measurement incompatibility as resources for nonlocality](#)
Defence date: 9 Sep 2016
Supervisor: [Nicolas Brunner](#)
Funding: [SNF](#), Switzerland

- Aug 2010 – **MSc in Physics**, UFMG, Belo Horizonte, Brazil
 Sep 2012 Thesis: **Black box correlations: locality, noncontextuality, and convex polytopes**
 Defence date: 12 Sep 2012
 Supervisor: **Marcelo Terra Cunha**
 Co-supervisor: **Daniel Cavalcanti**
 Funding: **CAPES**, Brazil
- Feb 2007 – **BSc in Physics**, UFMG, Belo Horizonte, Brazil
 Aug 2010 Monograph: **Não-localidade como recurso para comunicação**
 Supervisor: **Marcelo Terra Cunha**
 Funding: **CNPq**, Brazil

Research funding and fellowships

- Sep 2025 – **Principal Investigator and project coordinator**, **ANR JCJC HOQO-KS (ANR-25-CE47-6396)**, Sorbonne University / LIP6, France
 Aug 2030 *Higher-Order Quantum Operations with Known States*; funding: EUR 358,660
- Jan 2025 – **Research mobility project**, **PHC Polonium (53077ZD / BPN/BFR/2024/1/00026)**, together with **Michał Studziński**, Sorbonne University / University of Gdańsk, France and Poland
 Dec 2026 *Efficient methods for storing and transforming quantum programs*; bilateral mobility funding for research visits
- 2024 **Principal Investigator**, **Tremplins nouveaux entrants & nouvelles entrantes**, Sorbonne University / LIP6, France
Higher-Order Quantum Operations with Known States (HOQO-KS); funding: EUR 5,000
- Oct 2023 – **Funding for PhD student**, **QuantEdu France doctoral contract (via PCQT)**, Sorbonne University / LIP6, France
 Sep 2026 *Higher-Order Quantum Operations: Foundations and Applications*; funding: one 36-month doctoral contract from the French National Quantum Plan
- 2020 **Postdoctoral research fellowship**, **ESQ Postdoc Fellowship Program**, Austrian Academy of Sciences / IQOQI Vienna, Austria/EU
Higher-Order Quantum Operations: Foundations and Applications; funding: EUR 124,506
- Nov 2016 – **Co-Investigator / Foreign JSPS Fellow**, **Grant-in-Aid for JSPS Fellows (KAKENHI 16F16769)**, with **Mio Murao (PI)**, The University of Tokyo, Japan
 Mar 2019 *Entanglement and nonlocality in quantum theory*; direct-cost funding: JPY 2,100,000
- Nov 2016 – **Postdoctoral research fellow**, **JSPS Postdoctoral Fellowship for Research in Japan (Standard)**, The University of Tokyo, Japan
 Nov 2018 Funding: JPY 8,888,000
- 2012 **Master's research stipend**, **CAPES Master's stipend**, UFMG, Brazil
Black box correlations: locality, noncontextuality, and convex polytopes; funding: BRL 29,100

Languages

Portuguese (native), English (fluent), French (fluent), Spanish (advanced), Japanese (intermediate)

Technical skills

Scientific computing: MATLAB and Mathematica (advanced); familiarity with Julia, Python, and Bash.

I use GitHub (<https://github.com/mtcq>) to share research-related computational code.

Publications

Summary: 65 research articles: 52 peer-reviewed journal publications and 13 preprints. Journal publications include 13 Phys. Rev. Lett., 3 Nat. Commun., 1 J. Math. Phys., 2 IEEE Trans. Inf. Theory, and 14 Quantum.

- More than 110 co-authors.
- Up-to-date citation metrics are available on [Google Scholar](#).
- All my scientific papers are available at [arXiv](#) and on my [personal website](#).
- All publications since I moved to France may also be found on [HAL](#).
- **A list with all my publications can be found at the end of this CV.**

Responsibilities

- Sep 2023 – **Deputy coordinator (responsable adjoint)**, Sorbonne University's [Master's programme in Quantum Information](#). Curriculum design, admissions, and master's thesis examinations.
- 2026 **NII 1st-call internship (Tokyo, Japan)**: selected and ranked LIP6 applicants.
- 2026 **QuanG2 committee member (Grenoble, France)**: Autumn 2026 PhD call.
- 2025 **NII internships, both calls (Tokyo, Japan)**: selected and ranked LIP6 applicants ([first](#); [second](#)).
- 2025 **EDITE (ED 130) doctoral-contract jury (Paris, France)**: Operations Research, Algorithms, Computing and Programming panel member; chair: Antoine Genitrini.
- 2024 **NII 2nd-call internship (Tokyo, Japan)**: selected and ranked LIP6 applicants.

Research supervision

Worked closely with undergraduate, master's, and PhD students from different backgrounds at several institutions.

- Completed the co-supervision of 2 PhD students
- Completed the co-supervision of 2 master's theses
- Completed the supervision of 8 master's internship projects
- Currently supervising 4 PhD students

PhD supervision

- Dec 2025 – **PhD supervision, Sorbonne University, Paris, France**
 - Present Student name: Polina Chernyshova
 - Co-supervisor: Mio Murao, The University of Tokyo (Tokyo, Japan)
 - Project title: Higher-order quantum computing with partial information
- Nov 2025 – **PhD supervision, Sorbonne University, Paris, France**
 - Present Student name: Mariia Kvashchuk
 - Co-supervisor: Akihito Soeda, NII (Tokyo, Japan)
 - Project title: Quantum communication and discrimination in the single-shot regime

- Apr 2025 – **PhD supervision**, *Sorbonne University*, Paris, France
 Present Student name: Lucas Porto
 Co-supervisor: Mattia Walschaers
 Project title: The relationship between quantum entanglement and Bell nonlocality
- Oct 2023 – **PhD supervision**, *Sorbonne University*, Paris, France
 Present Student name: Vanessa Brzić
 Project title: Higher-Order Quantum Operations: Foundations and Applications
- PhD co-supervision**
- Jun 2017 – **PhD co-supervision**, *UFMG*, Belo Horizonte, Brazil
 Jan 2022 Student name: Marcello Nery
 Main supervisor: Reinaldo Oliveira Vianna
 Thesis title: Non-classical common-cause and direct-cause
 PhD defended: 21 Jan 2022
- Jan 2018 – **PhD co-supervision**, *The University of Tokyo*, Tokyo, Japan
 Mar 2021 Student name: Qingxiuxiong Dong
 Main supervisor: Mio Murao
 Thesis title: Quantum algorithms for higher-order quantum transformations of universal unitary operations
 PhD awarded: 19 Mar 2021
- Hosting PhD secondments**
- Mar 2024 – **PhD secondment**, *Sorbonne University*, Paris, France
 Apr 2024 Student name: Ties Ohst
 Main supervisor: Otfried Gühne
 Project title: Quantum memory in causally ordered processes of universal unitary operations
- Master's thesis co-supervision**
- Jul 2018 – **MSc co-supervisor**, *The University of Tokyo*, Tokyo, Japan
 Jul 2020 Student name: Wataru Yokojima
 Main supervisor: Mio Murao
 Thesis title: Consequences of preserving reversibility in quantum superchannels
 MSc defended: Feb 2020
- Jun 2017 – **MSc co-supervisor**, *The University of Tokyo*, Tokyo, Japan
 Jan 2018 Student name: Qingxiuxiong Dong
 Main supervisor: Mio Murao
 Thesis title: Quantum implementability of maps and supermaps
 MSc defended: Feb 2018
- Master's internship supervision**
- Sep 2025 – **Master's internship supervisor**, *Sorbonne University*, Paris, France
 Aug 2026 Kristin Sundal Lien
 Project title: Entanglement and channel discrimination: sometimes less is more
- Feb 2025 – **Master's internship supervisor**, *Sorbonne University*, Paris, France
 Aug 2025 Loïc Divry-Kennedy
 Project title: Discriminating qubit states and channels from any finite group

- Mar 2025 – **Master’s internship supervisor**, *Sorbonne University*, Paris, France
 Sep 2025 Charbel Eid
 Project title: Optimal discrimination between quantum measurements and quantum instruments
- Feb 2024 – **Master’s internship supervisor**, *Sorbonne University*, Paris, France
 Mar 2024 Yuki Koizumi
 Project title: Equivalence between unitary estimation and deterministic port-based teleportation
- Jan 2024 – **Master’s internship supervisor**, *Sorbonne University*, Paris, France
 Apr 2024 Shijun Zhang
 Project title: Perfect discrimination of unitary operations when k calls are available
- Apr 2023 – **Master’s internship supervisor**, *Sorbonne University*, Paris, France
 Jun 2023 Shijun Zhang
 Project title: Perfect discrimination of unitary operations when k calls are available
- Apr 2023 – **Master’s internship supervisor**, *Sorbonne University*, Paris, France
 Jun 2023 Charbel Eid
 Project title: Optimal discrimination between quantum measurements and instruments
- Apr 2023 – **Master’s internship supervisor**, *Sorbonne University*, Paris, France
 Jun 2023 Ethan Obadia
 Project title: Quantum channel discrimination with partially entangled states

Participation in examination boards

I have served on 10 PhD and 2 MSc examination committees, including as an external reviewer.

Membership of PhD juries

- 25 Jun 2026 **Thesis reviewer**, *Center for Theoretical Physics PAS*, Warsaw, Poland
 Student name: Oskar Szymon Słowik
 Thesis title: Spectral gaps, t-designs and ϵ -nets in quantum computing
 Supervisors: Adam Sawicki and Oliver Reardon-Smith
- 19 Mar 2026 **Thesis reviewer**, *Unicamp*, Campinas, Brazil
 Student name: José Nogueira
 Thesis title: Postquantum correlations and their consequences: From exclusivity graphs to Prepare-and-Measure scenarios
 Supervisors: Rafael Rabelo and Marcelo Terra Cunha
- 29 Oct 2025 **Thesis reviewer**, *National University of Singapore*, Singapore, Singapore
 Student name: Peter Sidajaya
 Thesis title: Studies in quantum foundations
 Supervisor: Valerio Scarani
- 6 Oct 2025 **Thesis reviewer**, *Université Grenoble Alpes*, Grenoble, France
 Student name: Pierre Pocreau
 Thesis title: The use of entanglement in non-cooperative games, and the computational power of indefinite causal order
 Supervisors: Mehdi Mhalla and Alastair Abbott
- 16 May 2025 **Thesis reviewer**, *University of Siegen*, Siegen, Germany
 Student name: Carlos Belini de Gois
 Thesis title: Certifying quantum states and their properties through correlations, communication and tomography
 Supervisor: Otfried Gühne

- 17 Dec 2024 **Thesis reviewer**, *Unicamp*, Campinas, Brazil
Student name: Lucas Pollyceno
Thesis title: Multipartite Bell nonlocality and communication
Supervisor: Rafael Rabelo
- 23 Oct 2023 **Thesis reviewer**, *Department of Physics, Technical University of Denmark*, Copenhagen, Denmark
Student name: Carles Roch i Carceller
Thesis title: Quantum state discrimination with applications in contextuality and randomness certification
Supervisor: Jonatan Bohr Brask
- 12 Jun 2023 **Thesis reviewer**, *Université libre de Bruxelles*, Brussels, Belgium
Student name: Jef Pauwels
Thesis title: Entanglement and quantum communication between partially characterized devices
Supervisor: Stefano Pironio
- 10 Mar 2023 **Thesis reviewer**, *Unicamp*, Campinas, Brazil
Student name: Carlos Humberto de Souza Vieira
Thesis title: Exploring the role of entanglement in sets of behaviors from prepare-and-measure scenarios
Supervisor: Marcelo Terra Cunha
- 3 Feb 2022 **Member of the PhD defence board**, *Unicamp*, Campinas, Brazil
Student name: Roberto Dobal Baldijão
Thesis title: Quantum Darwinism and Contextuality
Supervisor: Marcelo Terra Cunha
- Membership of MSc juries**
- 31 Jan 2024 **Thesis reviewer**, *Unicamp*, Campinas, Brazil
Student name: Arthur Couto Rosa Dutra de Oliveira
Thesis title: Can KS-Contextuality Hide in a Crowd? Investigating state-independent contextuality in systems with multiple observers
Supervisor: Marcelo Terra Cunha
- 9 Apr 2021 **Thesis reviewer**, *Unicamp*, Campinas, Brazil
Student name: Lucas da S. Pollyceno
Thesis title: Novos critérios para o princípio da causalidade de informação
Supervisor: Marcelo Terra Cunha
- Individual thesis monitoring committees (CSI)**
- 8 Jul 2026 **External member of the thesis monitoring committee**, *Université Grenoble Alpes*, Grenoble, France
Student name: Amanda Maria Fonseca
PhD project: Characterising causal structures: classical, quantum, and relativistic causality
Supervisors: Cyril Branciard (director) and V. Vilasini (co-director)
- 16 Jun 2026 **External member of the thesis monitoring committee**, *Sorbonne University*, Paris, France
Student name: Youness Ayaita
PhD project: L'induction et la probabilité
Supervisor: Anouk Barberousse
- 15 Jun 2026 **External member of the thesis monitoring committee**, *Institut de Physique Théorique (IPhT)*, Gif-sur-Yvette, France
Student name: Mu-En Liu
PhD project: Tsirelson Inequalities in the Minimal Bell Scenario
Supervisor: Jean-Daniel Bancal

- 24 Sep 2025 **External member of the thesis monitoring committee**, *Université Grenoble Alpes*, Grenoble, France
 Student name: Amanda Maria Fonseca
 PhD project: Characterising causal structures: classical, quantum, and relativistic causality
 Supervisors: V. Vilasini (Inria, QINFO) and Cyril Branciard (CNRS, Institut Néel)
- 5 Sep 2025 **External member of the thesis monitoring committee**, *Sorbonne University*, Paris, France
 Student name: Youness Ayaita
 PhD project: Le raisonnement ampliatif : Expliquer le succès épistémique de la science
 Supervisor: Anouk Barberousse
- 7 May 2025 **External member of the thesis monitoring committee**, *Université Paris-Saclay*, Palaiseau, France
 Student name: Octave Mestoudjian
 PhD project: Sous-systèmes généralisés en informatique quantique
 Supervisors: Pablo Arrighi, Augustin Vanrietvelde, and Mathew Wilson
- 8 Oct 2024 **External member of the thesis monitoring committee**, *Sorbonne University*, Paris, France
 Student name: Youness Ayaita
 PhD project: Le raisonnement ampliatif
 Supervisor: Anouk Barberousse
- 8 Jul 2024 **External member of the thesis monitoring committee**, *Université Grenoble Alpes*, Grenoble, France
 Student name: Pierre Pocreau
 PhD project: Implications of causal indefiniteness for quantum communication
 Supervisors: Mehdi Mhalla and Alastair Abbott
- 1 Jul 2024 **External member of the thesis monitoring committee**, *Université Paris-Saclay*, Palaiseau, France
 Student name: Octave Mestoudjian
 PhD project: Sous-systèmes généralisés en informatique quantique
 Supervisors: Pablo Arrighi and Augustin Vanrietvelde
- 21 Jun 2023 **External member of the thesis monitoring committee**, *Université Grenoble Alpes*, Grenoble, France
 Student name: Pierre Pocreau
 PhD project: Implications of causal indefiniteness for quantum communication
 Supervisors: Mehdi Mhalla and Alastair Abbott

Teaching summary

A detailed list of my teaching experience is presented at the end of this CV.

- At least 121 h of lectures, plus 4 one-off lectures
- At least 308 h of tutorials, plus 6 teaching-assistant appointments
- Responsible for creating/reformulating the course “Quantum Circuits and Query Complexity” (previously called “Quantum Circuits and Logic Gates”) for Sorbonne University’s **Master’s programme in Quantum Information**.
- Responsible for creating/reformulating the course “QIIntro” for Sorbonne University’s **Master’s programme in Quantum Information**.

Brief description of recent teaching

Undergraduate level in computer science at Sorbonne University:

- Mathématiques Discrètes (TD/TP, 3x)
- Introduction à la calculabilité et à la décidabilité (TD/TP, 3x)

Master’s level in the Quantum Information programme at Sorbonne University:

- Quantum Circuits and Query Complexity (lectures and tutorials, 3x)
- QIIntro (lectures and tutorials, 2x)
- Quantum Kinematics for CS (tutorials, 1x)

- Théorie de Shannon classique et quantique (lectures and tutorials, 2x)
- Master's level (outside quantum information) at Sorbonne University:
- Analyse d'algorithmes et génération aléatoire (TD/TP, 1x)

Outreach and dissemination

During my JSPS postdoc in Japan, I joined the **JSPS Science Dialogue Program**, a project where I presented my research in an accessible way to Japanese high school students. I am also interested in popular science events.

- 21 Sep 2018 **JSPS Science Dialogue**, Suwa Seiryō High School (Nagano Prefecture), Japan
What do you mean by random? Randomness, Quantum Mechanics, and Bell Non-locality
- 5 Dec 2017 **JSPS Science Dialogue**, Takezono High School (Ibaraki Prefecture), Japan
What do you mean by random? Randomness, Quantum Mechanics, and Bell Non-locality
- 7 Jan 2013 **Science Slam Berlin**, SO36, Berlin, Germany
Does God play dice? Genuine randomness in nature

Reviewing activities

Referee for the journals: Quantum, PRL, PRX, PRX Quantum, PRA, Phys. Rev. Research, J. Math. Phys., NJP, J. Phys. A, npj Quantum Information, Optica.

I have also refereed submissions for conferences such as QIP, STOC, TQC, and AQIS.

Scientific conferences

- Organised 1 workshop
- Served on the programme committees of 8 conferences
- Lecturer at 1 summer school
- Invited to present talks at 14 conferences and workshops (two events were cancelled)
- 23 contributed talks accepted at 22 conferences, including 3 talks accepted at **QIP**
- 15 posters accepted at conferences

Organisation

- Dec 2023 **Japanese-French Quantum Information 2023 Workshop**, Tokyo, Japan

Programme committees

- Jun 2026 **Causalworlds 2026**, Grenoble, France
- Oct 2025 **YQIS25**, Barcelona, Spain
- Aug 2025 **AQIS25**, The University of Hong Kong (HKU), Hong Kong
- Jun–Jul 2025 **CEQIP2025**, Smolenice Castle, Slovakia
- Sep 2024 **Causalworlds 2024**, Waterloo, Canada
- Aug 2024 **AQIS24**, Sapporo, Japan
- Sep 2019 **YQIS2019**, Gdańsk/Sopot, Poland
- Sep 2018 **YQIS2018**, Vienna, Austria

Summer school

Aug–Sep 2026 **QALYPSO 2026: Quantum Optimisation Techniques & Physics of Computation**, Valletta, Malta

Lecturer: Semidefinite Programming as an Analytical and Computational Tool for Quantum Information

Invited talks

Sep 2026 **Workshop on the Structure of Quantum Theory**, Münster, Germany

Forthcoming invited talk; title to be announced

Aug 2026 **YQIS 2026**, Vienna, Austria

Invited talk: Optimal pure state cloning and transposition are complementary channels

Apr 2026 **851st WE-Heraeus Seminar: 90 Years of Einstein–Podolsky–Rosen Paradox**, Physikzentrum Bad Honnef, Germany

Invited talk: Understanding the relationship between measurement incompatibility, EPR steering, and Bell nonlocality

Feb 2026 **Quantum Correlations and Measurements**, Les Diablerets, Switzerland

Invited talk: Optimal pure state cloning and optimal pure state transposition are complementary channels

Jul 2025 **CEQIP**, Smolenice Castle, Slovakia

Invited tutorial: Quantum information processing via higher-order operations

Jun 2025 **FoQaCiA Workshop 2025**, Braga, Portugal

Talk: Can the quantum switch be deterministically simulated?

Apr 2025 **BQIT:25**, Bristol, UK

Talk: Can the quantum switch be deterministically simulated?

Apr 2022 **Tsirelson Memorial Workshop**, Vienna, Austria

Talk: Measurement incompatibility and Bell nonlocality: from 1985 to 2022

Mar 2020 **Second Workshop on Quantum Protocols**, Waseda University, Tokyo, Japan

Workshop cancelled due to the COVID-19 pandemic

Planned talk: Reversing Unknown Quantum Transformations: Universal Quantum Circuit for Inverting General Unitary Operations

Mar 2020 **The Order of Things (TOOT)**, Obergurgl, Austria

Conference cancelled due to the COVID-19 pandemic

Jan 2020 **Quantum Information Structure of Spacetime (QISS) 2020**, Hong Kong

Talk: Reversing unknown quantum transformations: A universal quantum circuit for inverting general unitary operations

Nov 2018 **Quantum Maiwar**, Brisbane, Australia

Talk: Semi-device-independent certification of indefinite causal order

Sep 2018 **Post-AQIS2018 Workshop**, Nagoya, Japan

Invited talk: Reversing unknown quantum transformations: A universal protocol for inverting general unitary operations

Aug 2017 **FXQi Workshop: Quantum Incompatibility**, Laach Lake, Germany

Talk: Genuine n -wise measurement incompatibility and device-independent certificates of incompatibility

Accepted talks

- Aug 2026 **Quantum Physics and Logic 2026 (QPL2026)**, Amsterdam, The Netherlands
Talk: Composable simultaneous purification: when all communication scenarios reduce to spatial correlations
- Jul 2025 **Quantum Physics and Logic 2025 (QPL2025)**, Varna, Bulgaria
Talk: One-to-one correspondence between deterministic port-based teleportation and unitary estimation
- Jul 2025 **Quantum Physics and Logic 2025 (QPL2025)**, Varna, Bulgaria
Talk: All incompatible measurements on qubits lead to multiparticle Bell nonlocality
- Feb 2025 **Quantum Information Processing 2025 (QIP2025)**, Raleigh, United States
Talk: One-to-one correspondence between deterministic port-based teleportation and unitary estimation
(presented by Satoshi Yoshida)
- Aug 2023 **Asian Quantum Information Science 2023 (AQIS2023)**, Seoul, South Korea
Talk: Simulating qubit correlations with classical communication
- Feb 2023 **Quantum Information Processing 2023 (QIP2023)**, Ghent, Belgium
Talk: Simulating qubit correlations with classical communication
(presented by Martin J. Renner)
- Sep 2021 **SFB BeyondC Autumn Workshop 2021**, Innsbruck, Austria
Talk: Universal protocols for transforming unitary quantum operations
- Jun 2021 **Quantum Physics and Logic 2021 (QPL2021)**, Gdańsk (online), Poland
Talk: Success-or-draw: A strategy allowing repeat-until-success in quantum computation
- Jan 2020 **Quantum Information Processing 2020 (QIP2020)**, Shenzhen, China
Talk: Adaptive circuits exponentially outperform parallel ones for universal unitary inversion
- Aug 2019 **Asian Quantum Information Science 2019 (AQIS2019)**, Seoul, South Korea
Talk: Semi-device-independent certification of indefinite causal order
- Sep 2018 **Asian Quantum Information Science 2018 (AQIS2018)**, Nagoya, Japan
Talk: Reversing unknown quantum transformations
- Jul 2018 **Modern Topics in Quantum Information**, Natal, Brazil
Talk: Reversing unknown quantum transformations
- May 2017 **36th Quantum Information Technology Symposium (QIT36)**, Kyoto, Japan
Talk: Super-activation of quantum steering
- Mar 2016 **Quantum Networks (FQXi)**, Barcelona, Spain
Talk: Entangled states with a local hidden-variable model for sequential measurements
- Dec 2015 **Quantum Correlations, Contextuality and All That... Again**, Natal, Brazil
Talk: Algorithmic construction of local hidden variable models for entangled quantum states
- Aug 2015 **V Quantum Information School and Workshop**, Paraty, Brazil
Talk: Joint measurability, EPR steering, and Bell nonlocality
- Aug 2014 **Asian Quantum Information Science 2014 (AQIS2014)**, Kyoto, Japan
Talk: Joint measurability, EPR steering, and Bell nonlocality
- Dec 2013 **Quantum Correlations, Contextuality and All That**, Natal, Brazil
Talk: Measurement Incompatibility in Quantum Mechanics

- Aug 2013 **IV Quantum Information School and Workshop**, Paraty, Brazil
Talk: Genuine Hidden Quantum Nonlocality
- Jun 2012 **Workshop on Quantum Correlations**, Natal, Brazil
Talk: All noncontextuality inequalities for the n-cycle scenario
- Dec 2011 **III Encontro temático do INCT-IQ**, Natal, Brazil
Talk: From the detection loophole to the transmission loophole
- Aug 2011 **III Quantum Information School and Workshop**, Paraty, Brazil
Talk: Maximal CHSH violations with low efficiency photodetection and homodyne measurements
- Poster presentations
- Sep 2021 **Time in quantum theory, ETH workshop**, Zurich (online), Switzerland
Poster: Universal quantum circuits for transforming unitary operations: exponential advantages with causally adaptive strategies and the power of indefinite causality
- Sep 2021 **Vienna Quantum Foundations Conference (VQF-CON 2021)**, Vienna, Austria
Poster: Universal quantum circuits for transforming unitary operations: exponential advantages with causally adaptive strategies and the power of indefinite causality
- Jun 2020 **Theory of Quantum Computation, Communication and Cryptography 2020 (TQC2020)**, Riga (online), Latvia
Poster: Adaptive circuits exponentially outperform parallel ones for universal unitary inversion
- Dec 2019 **Topical Conference on Quantum Communication and Security 2019 (TCQC2019)**, Kyoto, Japan
Poster: Reversing unknown quantum transformations: A universal quantum circuit for inverting general unitary operations
- Aug 2019 **Asian Quantum Information Science 2019 (AQIS2019)**, Seoul, South Korea
Poster: Distributed sampling, quantum communication witnesses, and measurement incompatibility
- Apr 2018 **International Conference on Challenges in Quantum Information Science (CQIS18)**, Tokyo, Japan
Poster: The Cost of Implementing Non-Completely Positive Linear Maps
- Jan 2014 **Quantum Information Processing 2014 (QIP2014)**, Barcelona, Spain
Poster: Genuine Hidden Quantum Nonlocality
- Aug 2013 **IV Quantum Information School and Workshop**, Paraty, Brazil
Poster: Realistic loophole-free Bell test with atom-photon entanglement
- Jan 2013 **Quantum Information Processing 2013 (QIP2013)**, Beijing, China
Poster: Towards a loophole-free Bell test with continuous-variable systems
- May 2012 **TQC2012**, Tokyo, Japan
Poster: Perfect homodyne measurements imply CHSH violation with arbitrarily low photodetection efficiency
- Nov 2010 **XIX Semana da Iniciação Científica, UFMG**, Belo Horizonte, Brazil
Poster: Jogo do Quadrado Mágico; Pseudotelepatia Quântica
- Nov 2010 **V Simpósio Nacional / Jornadas de Iniciação Científica, IMPA**, Rio de Janeiro, Brazil
Poster: Não-localidade como recurso para comunicação

- Oct 2010 **WECIQ2010**, Petrópolis, Brazil
Poster: Jogo do Quadrado Mágico; Pseudotelepatia Quântica
- Oct 2009 **XVIII Semana da Iniciação Científica**, UFMG, Belo Horizonte, Brazil
Poster: Algoritmo de Grover – Selected among the top 8%
- Oct 2008 **XVII Semana da Iniciação Científica**, UFMG, Belo Horizonte, Brazil
Poster: Números Inteiros e Criptografia RSA

Participation

- Jun 2025 **Quantum Information (Benasque)**, Benasque, Spain
Workshop without formal talks
- 15–19 Apr 2024 **Foundations of Quantum Physics beyond Bell: Celebrating 60 years of Bell's theorem**, Les Diablerets, Switzerland
- Jun 2023 **Quantum Information (Benasque)**, Benasque, Spain
Workshop without formal talks
- Jun 2019 **Quantum Information (Benasque)**, Benasque, Spain
Workshop without formal talks
- Jun 2017 **Quantum Information (Benasque)**, Benasque, Spain
Workshop without formal talks
- Jun 2015 **Quantum Information (Benasque)**, Benasque, Spain
Workshop without formal talks
- Aug 2015 **V Quantum Information School and Workshop**, Paraty, Brazil
- Sep 2015 **Gisin's 60th birthday workshop**, Riederalp, Switzerland
- Jun 2013 **Quantum Information (Benasque)**, Benasque, Spain
Workshop without formal talks
- Aug 2013 **IV Quantum Information School**, Paraty, Brazil
- Jul 2012 **62nd Lindau Nobel Laureate Meeting dedicated to Physics**, Lindau, Germany
- Aug 2011 **III Quantum Information School and Workshop**, Paraty, Brazil
- Jul 2011 **28º Colóquio Brasileiro de Matemática**, IMPA, Rio de Janeiro, Brazil
- Aug 2010 **XIV Escola Brasileira de Probabilidade**, Búzios, Brazil
- Jul 2010 **Clay Mathematics Institute 2010 Summer School**, *Probability and Statistical Physics in Two and More Dimensions*, Búzios, Brazil

Academic visit seminars

- 30 Apr 2025 **University of Siegen**, Siegen, Germany
Talk: Can the quantum switch be deterministically simulated?
Host: Otfried Gühne
- 12 Dec 2024 **The University of Hong Kong (HKU)**, Hong Kong
Talk: Can the quantum switch be deterministically simulated?
Host: Giulio Chiribella
- 11 Nov 2024 **University of Geneva**, Geneva, Switzerland
Talk: All incompatible measurements on qubits lead to multiparticle Bell nonlocality
Host: Nicolas Brunner

- 8 Jul 2024 **Université Grenoble Alpes**, Grenoble, France
Talk: All incompatible measurements on qubits lead to multiparticle Bell nonlocality
Host: Alastair Abbott
- 25 Jun 2024 **University of Pisa**, Pisa, Italy
Talk: The relationship between measurement incompatibility and Bell nonlocality
Host: Costantino Budroni
- 24 Oct 2023 **Technical University of Denmark, QPIT**, Copenhagen, Denmark
Talk: Transforming and discriminating quantum operations using higher-order methods
Host: Jonatan Bohr Brask
- 16 Mar 2023 **Les Ateliers du LKB, Laboratoire Kastler Brossel**, Paris, France
Talk: Parallel, sequential, and non-causal strategies for transforming unitary operations and discriminating quantum channels via a higher-order approach.
Host: Nancy Paul
- 10 Dec 2021 **University of Gdańsk**, Gdańsk, Poland
Talk: Transforming unitary operations via quantum circuits: Universal unitary inversion, transposition, and complex conjugation
Host: Michał Studziński
- 18 Mar 2021 **Quantin Research Group**, Warsaw (online), Poland
Talk: Universal protocols for transforming unitary quantum operations
Host: Michał Oszmaniec
- 5 Feb 2021 **Technical University of Denmark**, Copenhagen (online), Denmark
Talk: Reversing unknown quantum transformations: A universal protocol for inverting general unitary operations
Host: Jonatan Bohr Brask
- 10 Jul 2019 **Universitat Autònoma de Barcelona**, Barcelona, Spain
Talk: Reversing unknown quantum transformations: A universal protocol for inverting general unitary operations
Host: Andreas Winter
- 14 Feb 2019 **ICFO**, Barcelona, Spain
Talk: Reversing unknown quantum transformations: A universal protocol for inverting general unitary operations
Host: Antonio Ácin
- 12 Feb 2019 **GAP**, Geneva, Switzerland
Talk: Reversing unknown quantum transformations: A universal protocol for inverting general unitary operations
Host: Nicolas Brunner
- 24 Jan 2019 **IQOQI Vienna**, Vienna, Austria
Talk: Reversing unknown quantum transformations: A universal protocol for inverting general unitary operations
Host: Marcus Huber
- 16 Dec 2016 **The University of Tokyo**, Tokyo, Japan
Talk: Super-Activation of Quantum Steering
Host: Mio Murao

- 18 Nov 2016 **UFMG**, Belo Horizonte, Brazil
Talk: Super-Activation of Quantum Steering
Host: Marcelo Terra Cunha
- 18 Nov 2015 **University of Siegen**, Siegen, Germany
Talk: Inequivalence of Entanglement, Steering, and Bell Nonlocality For General Measurements
Host: Otfried Gühne
- 18 Aug 2015 **UFMG**, Belo Horizonte, Brazil
Talk: Inequivalence of Entanglement, Steering, and Bell Nonlocality For General Measurements
Host: Marcelo Terra Cunha
- 31 Jul 2015 **UFMG**, Belo Horizonte, Brazil
Talk: Joint Measurability, EPR Steering, and Bell Nonlocality
Host: Marcelo Terra Cunha
- 15 Jun 2015 **Waseda University**, Tokyo, Japan
Talk: Joint Measurability, EPR Steering, and Bell Nonlocality
Host: Kazuya Yuasa
- 26 May 2015 **Perimeter Institute (PI)**, Waterloo, Canada
Talk: Joint Measurability, EPR Steering, and Bell Nonlocality
Host: Matthew Pusey
- 9 Oct 2014 **ICFO**, Barcelona, Spain
Talk: Joint Measurability, EPR Steering, and Bell Nonlocality
Host: Antonio Ácin
- 5 Sep 2014 **KCIK**, Gdańsk, Poland
Talk: Joint Measurability, EPR Steering, and Bell Nonlocality
Host: Michał Horodecki
- 25 Aug 2014 **The University of Tokyo**, Tokyo, Japan
Talk: Joint Measurability, EPR Steering, and Bell Nonlocality
Host: Mio Murao
- 3 Dec 2013 **UFMG**, Belo Horizonte, Brazil
Talk: Genuine Hidden Quantum Nonlocality
Host: Marcelo Terra Cunha
- 19 Jul 2012 **KCIK**, Gdańsk, Poland
Talk: Characterization of the n-Cycle Noncontextual Polytope
Host: Michał Horodecki
- 11 Jul 2012 **ICFO**, Barcelona, Spain
Talk: Characterization of the n-Cycle Noncontextual Polytope
Host: Antonio Ácin

List of publications

Peer-reviewed publications

- [1] L. E. A. Porto, S. Designolle, S. Pokutta, and M. T. Quintino. "Measurement incompatibility and quantum steering via linear programming". *Quantum* **10**, 2141 (2026).

- [2] F. Grosshans, M. Horodecki, M. Murao, T. Młynik, M. T. Quintino, M. Studziński, and S. Yoshida. “Multicopy quantum state teleportation with application to storage and retrieval of quantum programs”. *Quantum* **10**, 2105 (2026).
- [3] S. Yoshida, Y. Koizumi, M. Studziński, M. T. Quintino, and M. Murao. “One-to-one correspondence between deterministic port-based teleportation and unitary estimation”. *IEEE Trans. Inf. Theory* **72**, 2358–2377 (2026).
- [4] T.-A. Ohst, S. Zhang, H. C. Nguyen, M. Plávala, and M. T. Quintino. “Characterising memory in quantum channel discrimination via constrained separability problems”. *Quantum* **10**, 1988 (2026).
- [5] J. Bavaresco, H. Kristjánsson, M. Murao, T. Otake, M. T. Quintino, P. Taranto, and S. Yoshida. “Simulating the quantum switch with quantum circuits is computationally hard”. *Nat. Commun.* **16**, 10216 (2025).
- [6] M. Plávala, O. Gühne, and M. T. Quintino. “All incompatible measurements on qubits lead to multiparticle Bell nonlocality”. *Phys. Rev. Lett.* **134**, 200201 (2025).
- [7] S. Egelhaaf, J. Pauwels, M. T. Quintino, and R. Uola. “Certifying measurement incompatibility in prepare-and-measure and Bell scenarios”. *J. Phys. A: Math. Theor.* **58**, 095304 (2025).
- [8] S. Milz and M. T. Quintino. “Characterising transformations between quantum objects, ‘completeness’ of quantum properties, and transformations without a fixed causal order”. *Quantum* **8**, 1415 (2024).
- [9] P. Taranto, M. T. Quintino, M. Murao, and S. Milz. “Characterising the hierarchy of multi-time quantum processes with classical memory”. *Quantum* **8**, 1328 (2024).
- [10] T. Strömberg, P. Schiainsky, M. T. Quintino, M. Antesberger, L. A. Rozema, I. Agresti, Č. Brukner, and P. Walther. “Experimental superposition of a quantum evolution with its time reverse”. *Phys. Rev. Research* **6**, 023071 (2024).
- [11] L. Villegas-Aguilar, E. Polino, F. Ghafari, M. T. Quintino, K. T. Laverick, I. R. Berkman, S. Rogge, L. K. Shalm, N. Tischler, E. G. Cavalcanti, S. Slussarenko, and G. J. Pryde. “Nonlocality activation in a photonic quantum network”. *Nat. Commun.* **15**, 3112 (2024).
- [12] M. Antesberger, M. T. Quintino, P. Walther, and L. A. Rozema. “Higher-order process matrix tomography of a passively-stable quantum switch”. *PRX Quantum* **5**, 010325 (2024).
- [13] Q. Dong, M. T. Quintino, A. Soeda, and M. Murao. “The quantum switch is uniquely defined by its action on unitary operations”. *Quantum* **7**, 1169 (2023).
- [14] M. Renner and M. T. Quintino. “The minimal communication cost for simulating entangled qubits”. *Quantum* **7**, 1149 (2023).
- [15] T. Strömberg, P. Schiainsky, R. Peterson, M. T. Quintino, and P. Walther. “Demonstration of a quantum SWITCH in a Sagnac configuration”. *Phys. Rev. Lett.* **131**, 060803 (2023).
- [16] D. Ebler, M. Horodecki, M. Marciniak, T. Młynik, M. T. Quintino, and M. Studziński. “Optimal universal quantum circuits for unitary complex conjugation”. *IEEE Trans. Inf. Theory* **69**, 5069–5082 (2023).
- [17] E.-C. Boghiu, F. Hirsch, P.-S. Lin, M. T. Quintino, and J. Bowles. “Device-independent and semi-device-independent entanglement certification in broadcast Bell scenarios”. *SciPost Phys. Core* **6**, 028 (2023).

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- [21] M. T. Quintino and D. Ebler. “Deterministic transformations between unitary operations: Exponential advantage with adaptive quantum circuits and the power of indefinite causality”. *Quantum* **6**, 679 (2022).
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- [30] L. Guerini, M. T. Quintino, and L. Aolita. “Distributed sampling, quantum communication witnesses, and measurement incompatibility”. *Phys. Rev. A* **100**, 042308 (2019).
- [31] M. T. Quintino, C. Budroni, E. Woodhead, A. Cabello, and D. Cavalcanti. “Device-independent tests of structures of measurement incompatibility”. *Phys. Rev. Lett.* **123**, 180401 (2019).
- [32] J. Bavaresco, M. Araújo, Č. Brukner, and M. T. Quintino. “Semi-device-independent certification of indefinite causal order”. *Quantum* **3**, 176 (2019).
- [33] Q. Dong, M. T. Quintino, A. Soeda, and M. Murao. “Implementing positive maps with multiple copies of an input state”. *Phys. Rev. A* **99**, 052352 (2019).
- [34] F. Hirsch, M. T. Quintino, and N. Brunner. “Quantum measurement incompatibility does not imply Bell nonlocality”. *Phys. Rev. A* **97**, 012129 (2018).

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- [48] F. Hirsch, M. T. Quintino, J. Bowles, and N. Brunner. “Genuine hidden quantum nonlocality”. *Phys. Rev. Lett.* **111**, 160402 (2013).
- [49] M. Araújo, M. T. Quintino, C. Budroni, M. Cunha, and A. Cabello. “All noncontextuality inequalities for the n -cycle scenario”. *Phys. Rev. A* **88**, 022118 (2013).
- [50] C. Teo, M. Araújo, M. T. Quintino, J. Minář, D. Cavalcanti, V. Scarani, M. Terra Cunha, and M. França Santos. “Realistic loophole-free Bell test with atom-photon entanglement”. *Nat. Commun.* **4**, 2104 (2013).
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- [53] F. Costa, G. Rubino, C. Branciard, Č. Brukner, and M. T. Quintino. “Indefinite Quantum Causality”. *arXiv e-prints* (2026). arXiv:2606.19438 [quant-ph].
- [54] K. S. Lien and M. T. Quintino. “Entanglement in quantum channel discrimination: sometimes less is more”. *arXiv e-prints* (2026). arXiv:2605.31519 [quant-ph].
- [55] M. Kвашчук, P. Chernyshova, L. E. A. Porto, T.-A. Ohst, L. B. Vieira, and M. T. Quintino. “The most discriminable quantum states in the multicopy regime”. *arXiv e-prints* (2026). arXiv:2604.26927 [quant-ph].
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- [57] V. Brzić, D. Grinko, M. Studziński, and M. T. Quintino. “Optimal pure state cloning and transposition are complementary channels”. *arXiv e-prints* (2026). arXiv:2603.23628 [quant-ph].
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- [65] A. Cabello, M. T. Quintino, and M. Kleinmann. “Possible consequences for physics of the negative resolution of Tsirelson’s problem”. *arXiv e-prints* (2023). arXiv:2307.02920 [quant-ph].

Theses and monographs

- [66] M. T. Quintino. *Quantum entanglement and measurement incompatibility as resources for nonlocality*. PhD thesis. 2016. URL: <https://archive-ouverte.unige.ch/unige:88093>.
- [67] M. T. Quintino. *Black box correlations: locality, noncontextuality, and convex polytopes*. MSc thesis. 2012. URL: <https://repositorio.ufmg.br/handle/1843/BUOS-A46HJC>.
- [68] M. T. Quintino. *Não-localidade como recurso para comunicação*. BSc monograph presented at IMPA. 2010. URL: <https://www.ime.unicamp.br/~tcunha/MonografiaMTulio.pdf>.

Other publications

- [69] M. T. Quintino. “Quantum teleportation beyond its standard form: Multi-Port-Based Teleportation”. *Quantum Views* **5**, 56 (2021).

Teaching details

- 2025/2 **Lecture (20 h), Tutorial (6 h)**, Sorbonne University, Master d’informatique - Information Quantique (IQ), M1, Paris, France
MU4INQ05 - Quantum Circuits and Query Complexity
- 2025/1 **Lecture (20 h)**, Sorbonne University, Master d’informatique - Information Quantique (IQ), M1, Paris, France
MU4INQ51 - QIIntro
- 2025/1 **Tutorial (36.5 h)**, Sorbonne University, Licence d’informatique, L3, Paris, France
LU3IN030 - Introduction à la calculabilité et à la décidabilité
- 2024/2 **Lecture (4 h), Tutorial (2 h)**, Sorbonne University, Master d’informatique - Information Quantique (IQ), M2, Paris, France
MU5PYQ03 - QIT: Quantum Information Theory
- 2024/2 **Lecture (10 h), Tutorial (8 h)**, Sorbonne University, Master d’informatique - Information Quantique (IQ), M1, Paris, France
MU4INQ05 - Quantum Circuits and Logic Gates
- 2024/2 **Tutorial (38.5 h)**, Sorbonne University, Licence d’informatique, L2, Paris, France
LU2IN005 - Mathématiques Discrètes
- 2024/1 **Lecture (2 h)**, *4EU+ European University Alliance*, University of Copenhagen (online), Denmark
4EU+: Quantum Information and Quantum Many-Body Theory
- 2024/1 **Lecture (10 h), Tutorial (6 h)**, Sorbonne University, Master d’informatique - Information Quantique (IQ), M1, Paris, France
MU4INQ51 - QIIntro
- 2024/1 **Lecture (18 h), Tutorial (6 h)**, Sorbonne University, Master Sciences, Technologies, Santé, M1, Paris, France
MU4PY223 - Théorie de Shannon classique et quantique
- 2024/1 **Tutorial (36.5 h)**, Sorbonne University, Licence d’informatique, L3, Paris, France
LU3IN030 - Introduction à la calculabilité et à la décidabilité
- 2023/2 **Lecture (4 h), Tutorial (2 h)**, Sorbonne University, Master d’informatique - Information Quantique (IQ), M2, Paris, France
MU5PYQ03 - QIT: Quantum Information Theory
- 2023/2 **Lecture (10 h), Tutorial (6 h)**, Sorbonne University, Master d’informatique - Information Quantique (IQ), M1, Paris, France
MU4INQ05 - Quantum Circuits and Logic Gates
- 2023/2 **Tutorial (38.5 h)**, Sorbonne University, Licence d’informatique, L2, Paris, France
LU2IN005 - Mathématiques Discrètes

- 2023/1 **Lecture (15 h), Tutorial (15 h)**, *Sorbonne University, Master Sciences, Technologies, Santé, M1*, Paris, France
MU4PY223 - Théorie de Shannon classique et quantique
- 2023/1 **Tutorial (36.5 h)**, *Sorbonne University, Licence d'informatique, L3*, Paris, France
LU3IN030 - Introduction à la calculabilité et à la décidabilité
- 2022/2 **Tutorial (36.5 h)**, *Sorbonne University, Licence d'informatique, L2*, Paris, France
LU2IN005 - Mathématiques Discrètes
- 2022/2 **Lecture (2 h), Tutorial (16 h)**, *Sorbonne University, Master d'informatique - Information Quantique (IQ), M1*, Paris, France
MU4INQ01 - Quantum Kinematics for Computer Scientists
- 2022/2 **Tutorial (14 h)**, *Sorbonne University, Master d'informatique - Parcours Science et Technologie du Logiciel (STL), M2*, Paris, France
MU5IN550 - Analyse d'algorithmes et génération aléatoire
- 2022/2 **Lecture (6 h), Tutorial (4 h)**, *Sorbonne University, Master d'informatique - Information Quantique (IQ), M2*, Paris, France
MU5PYQ03 - QIT: Quantum Information Theory
- 2021 **One lecture**, *Semana da Física 2021 – UNESP*, Guaratinguetá (online), Brazil
Não-localidade de Bell: como o indeterminismo quântico permite correlações supra-clássicas
- 2019 **One lecture**, *The University of Tokyo*, Tokyo, Japan
What do you mean by random? Randomness, Quantum Mechanics, and Bell Non-locality – Undergraduate course in quantum technology
- 2016/1 **Teaching Assistant**, *Université de Genève*, Geneva, Switzerland
Quantum Information Theory – master's course
- 2015/2 **Teaching Assistant**, *Université de Genève*, Geneva, Switzerland
Méthodes mathématiques pour physiciens I – undergraduate course
- 2015/1 **Teaching Assistant**, *Université de Genève*, Geneva, Switzerland
Quantum Information Theory – master's course
- 2014/1 **Teaching Assistant**, *Université de Genève*, Geneva, Switzerland
Quantum Information Theory – master's course
- 2013/1 **Teaching Assistant**, *Université de Genève*, Geneva, Switzerland
Quantum Information Theory – master's course
- 2011/1 **Teaching Assistant**, *IMPA*, Rio de Janeiro, Brazil
Mecânica quântica para matemáticos em formação – 28^o Colóquio Brasileiro de Matemática
- 2011 **One lecture**, *UFMG*, Belo Horizonte, Brazil
Comunicação via qubits – XXII Escola de Inverno
- 2010 **One lecture**, *UFMG*, Belo Horizonte, Brazil
Desigualdades de Bell, uma introdução à não-localidade quântica – Quantum Mechanics Graduate Course